

Edgar Ontiveros

Machine Learning Engineer | Developer



Professional Profile

Developer and machine learning engineer with a Physics Engineering background. I take systems from idea to production: I design the API in FastAPI, model the data in PostgreSQL, build the interface in React, and deploy on AWS with Docker and CI/CD. Machine learning goes in where it solves something real — forecasting, classification, similarity matching, CNNs — always served as an application people use daily.

Contact

Edgar Ontiveros

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Technical Stack

Backend & APIs: Python · FastAPI · Django · REST APIs · Pydantic · SQLAlchemy · SQLModel · Alembic · WSGI / ASGI · Uvicorn

Databases: PostgreSQL · SQL · SQLite · Supabase · Data Modeling · Migrations

Cloud & DevOps: AWS (EC2, EBS) · Docker · Docker Compose · GitHub Actions · CI/CD · nginx · Linux

Machine Learning: Scikit-learn · Forecasting · Classification · Regression · Deep Learning / CNNs · Model Evaluation

Data & ETL: Pandas · NumPy · ETL Pipelines · Parquet · Data Validation

Frontend: React · TypeScript · Vite · Streamlit

Testing & Tools: pytest · ruff · mypy · Git · GitHub · VS Code · Jupyter

BI & ERP: Power BI · DAX · Power Query · SAP · QAD · MRP

Languages

Spanish (Native)
English — B2

EDUCATION

2019 – 2024

B.Sc. in Physics Engineering

GRADUATE · Autonomous University of Chihuahua, Faculty of Engineering 🏛️

ACHIEVEMENTS

- Presenter, 67th National Physics Congress (2024)
- Participant, 66th National Physics Congress (2023)
- 1st Summer of Scientific Research (2023)
- Vice President, Student Council UACH

PROFESSIONAL EXPERIENCE

Apr 2026 – Present **Data Scientist / ML Engineer**

HERINOX · Mexico 📍

Build and deploy the internal software platform — APIs, web applications and machine learning services — used across the company's 11 branches.

- Design and ship full-stack internal applications with FastAPI, SQLAlchemy and PostgreSQL on the backend and React + TypeScript on the frontend, replacing manual Excel workflows with self-service tools for the sales, purchasing and pricing teams.
- Deploy and operate these services on AWS EC2 using Docker, nginx and GitHub Actions pipelines that run tests, linting and type checks before a gated production release.
- Model and migrate relational schemas in PostgreSQL (SQLAlchemy 2.0, Alembic), including business-hours SLA tracking across multiple time zones.
- Apply machine learning to internal operations — scikit-learn and fuzzy matching to auto-generate SAP article codes and classify products from free-text descriptions, plus forecasting models for sales, demand and inventory across all 11 branches.

Sep 2025 – Apr 2026 **Manufacturing Engineer 4.0**

SAFRAN AEROSYSTEMS · Chihuahua, MX 📍

- Built Python monitoring for automated production lines, processing live machine data to flag anomalies before they reached scrap.
- Led an electric-torque automation project that replaced a manual tightening operation, digitizing traceability and reducing torque-related defects.

Nov 2024 – Jul 2025 **Project Engineer Intern**

SAFRAN AEROSYSTEMS · Chihuahua, MX 📍

- Supported the industrial transfer of production lines from Niort, France, standardizing processes and automation for the new site.

PROJECTS

- **Quotation Management System** – FastAPI · PostgreSQL 17 · React 19 / TypeScript · Docker · GitHub Actions. Internal platform for 11 branches: quote requests, business-hours SLA tracking across time zones, quoted vs. confirmed revenue. Alembic migrations, background scheduler, pytest against real PostgreSQL, CI/CD with gated deploys. 🔄 proyecto-cotizaciones
- **Automated Pricing Engine** – FastAPI · React · Docker · Pandas. Turns SAP master data into branch price lists, cost sheets and variance reports under supplier-priority rules and fail-closed validation. Migrated from Streamlit to FastAPI + React, keeping the original as a parity test. 🔄 auto-precios
- **SAP Article Code Generator** – FastAPI · Scikit-learn · RapidFuzz · Docker · AWS EC2/EBS. Infers SAP code, product hierarchy, unit of measure and weight for new items from the most similar historical records. Multi-stage Docker build on EC2 with persistent EBS storage. 🔄 Generador-de-Codigos
- **Purchase Order Processor** – FastAPI · Pydantic · lxml · pdfplumber · Docker. Parses CFDI XML invoices and PDF proformas into SAP-ready Excel, converting measurement units and grouping line items. Layered architecture, supplier-parser Protocol, strict mypy, 97 tests. 🔄 Ordenes-Compra
- **CNN for Skin Cancer Detection** – TensorFlow/Keras · VGG16. Transfer-learning classifier with preprocessing and augmentation; 88% F1-score.

VOLUNTEER

- Mathematics & Physics Mentor, UACH outreach programmes (2019 – 2024)

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